Novelty Assessment:
Delay–Layout Coupling in UWB Self-Calibration

Prior-Art and Identifiability Analysis (Paper B: Mechanism)

Zekai Xiao

June 2026

This is the mechanism-paper novelty assessment (Paper B, an identifiability/error-analysis paper for a metrology venue); its companion is the AutoPos system-paper assessment (Paper A). The split is deliberate and non-overlapping: Paper A reports the working, ground-truth-validated system; Paper B explains why *its residual vertical error exists and why added geometry cannot remove it.*

1 One-Sentence Thesis

UWB anchor self-calibration couples antenna-delay/range-bias parameters with layout scale, which makes same-environment tag accuracy unreliable as a calibration-quality metric. In the Erlangen campaign the `v4-io` layout is scale-biased (metric-incorrect): an under-absorbed common-mode range bias expands it to an AutoPos-to-Vicon Sim(3) scale of 0.958—this is a range-bias signature, not a true geometric distortion. This scale-biased layout gives a better static median (72.7 mm) than the metric-correct common-mode layout (109.5 mm) *only when the tag delay is uncorrected*, because the layout's scale bias compensates an uncorrected positive tag-device delay. The effect is conditional: once the tag delay is corrected with a single global constant the metric-correct layout becomes the better positioner (68.5 mm held-out median, P95 156 mm, versus the scale-biased layout's 72.7 mm; better at 17 of 24 positions, paired $p \approx 0.02$, though the marginal-median gap alone is within noise), so the claim is not that a metric-incorrect layout is inherently a better positioner, but that same-environment accuracy can rank a scale-biased layout above an independently validated metric-correct one when a correctable tag/range bias is left uncorrected.

2 What the Full Text Confirms

2.1 UWB Self-Calibration

The UWB self-calibration literature already treats anchor deployment as a major cost and addresses automatic anchor localization from radio measurements, robot trajectories, multihop constraints, or auxiliary sensing. Hamer and D'Andrea derive a self-calibrating UWB network for multi-robot localization, including clock synchronization and anchor position computation (Hamer & D'Andrea, 2018). Ledergerber, Hamer, and D'Andrea show a one-way UWB robot self-localization system (Ledergerber et al., 2015). Almansa et al. study mobile UWB autocalibration for ad-hoc robot deployments, while Corbalan et al. evaluate large-scale UWB anchor self-localization in practical multihop testbeds (Almansa et al., 2020; Corbalan et al., 2023). Van

Herbruggen et al. and related self-calibration work address multihop or collaborative anchor positioning (Van Herbruggen et al., 2023; Ridolfi et al., 2021; Shi et al., 2019). These papers are close in deployment motivation, but their success metric is usually localization or anchor reconstruction after calibration. They do not report the paradox that a scale-biased layout can outperform a metric-correct one because of tag-side NLOS cancellation.

Lutz et al. are the closest methodological prior art because their full text explicitly calibrates both UWB anchor poses and a UWB sensor error model using visual-inertial SLAM and fiducial markers (Lutz et al., 2019). Their problem formulation acknowledges that radio propagation makes a consistent sensor error model difficult across anchor setups. That is close to our concern, but the estimator is aided by VINS/AprilTags and does not expose the scale–delay coupling of range-only anchor self-calibration or the in-environment cancellation that lets the scale-biased layout outperform the metric-correct one.

2.2 Antenna Delay, Range Bias, and Error Models

The antenna-delay and range-bias literature is extensive and directly relevant. Shalaby et al. propose a robust antenna-delay calibration procedure and model bias/uncertainty as a function of received-signal power (Shalaby et al., 2023). Ledergerber and D'Andrea calibrate module-dependent delays and range-error models, including a maximum-likelihood approach that does not require external ground truth (Ledergerber & D'Andrea, 2018). De Preter et al. jointly model range bias and anchor autocalibration for a UWB positioning system (De Preter et al., 2019); this is the closest UWB prior art and must be compared against the full text. By its formulation it estimates anchor geometry together with a constant range-bias model, but it does not appear to validate the recovered geometry against an independent metric reference—the separation this paper adds. Piavanini et al. estimate antenna delay and anchor self-localization in a common calibration workflow (Piavanini et al., 2022). Liu et al. and Shah et al. further show that antenna delays and node-specific ranging effects are important calibration targets (Liu et al., 2024; Shah et al., 2021, 2022).

These papers establish that delay and bias calibration matters. In UWB two-way-ranging work delay/range bias is typically estimated as a nuisance correction; in the acoustic/TOA self-calibration literature, by contrast, delay and offset are already first-class jointly-solved unknowns (see the Cross-Domain subsection). The closest UWB bias-calibration work is Yogesh et al. (2024), which estimates per-node additive range bias in a fully-connected UWB swarm validated against optical ground truth, but *explicitly* models the bias as pure-additive with no gain/scale component—a modelling choice their *known* node positions permit. Our result is the complementary one: once the node positions are unknown (self-calibration), that same additive bias shares a weakly observable direction with layout scale—the very coupling their known-position setting assumes away—so same-environment position error can prefer a metric-incorrect (scale-biased) calibration when the bias is left uncorrected.

2.3 NLOS Detection and Mitigation

The NLOS literature already shows that UWB ranges are often positively biased and that features from channel statistics, received signal properties, or learned models can mitigate those errors (Marano et al., 2010; Wymeersch et al., 2012; Guvenc et al., 2007; Venkatesh & Buehrer, 2007; Meghani et al., 2019; Jeong et al., 2023; Kong, 2023; Sung et al., 2023; Yang et al., 2024). Di Franco et al. are particularly relevant because they treat biased, multimodal NLOS UWB measurements in a calibration-free network-localization setting (Di Franco et al., 2017). However, these works aim to classify or correct NLOS, not to use NLOS reduction as a causal intervention that tests which calibration geometry is physically correct.

The Erlangen mechanism should therefore not be framed as a new NLOS classifier. In this campaign the scale-biased `v4-io` layout gives a better same-environment static median (72.7 mm) than the metric-correct common-mode layout (109.5 mm) when the tag delay is left uncorrected, because the layout's scale bias—an under-absorbed common-mode range bias appearing as expansion—compensates an uncorrected positive tag-device delay; per-frame joint tag-delay estimation restores the common-mode layout to 72.9 mm but destabilizes the tail (P95 317.5 mm). A raw-frame intervention that reduces the positive tag-anchor range tail reverses the in-environment ranking in favour of the metric-correct layout; this causal intervention is the decisive evidence for the cancellation mechanism.

2.4 Identifiability and Observability

The broader localization literature already contains the mathematical language needed to discuss weak directions. Aspnes et al. connect network localization to graph rigidity and unique localizability (Aspnes et al., 2006). Patwari et al. survey cooperative localization and statistical measurement models for wireless sensor networks (Patwari et al., 2005). Wymeersch et al. and Shen/Win-style work analyze cooperative localization, position-error bounds, geometry, and bias (Wymeersch et al., 2009; Jourdan et al., 2008). Hesch et al. and Goudar et al. show how observability and calibration consistency matter in visual-inertial or UWB-inertial systems (Hesch et al., 2014; Goudar et al., 2021).

Prorok's thesis (Prorok, 2013) derives a Cramér–Rao lower bound for the TDOA measurement model parameters $\theta = [\mu_u, \sigma_u, \mu_v, \sigma_v, P_{L,u}, P_{L,v}]^T$, which are the statistical parameters of the NLOS error model given *known* base-station positions. This is a positioning-side CRLB, not a calibration-side identifiability analysis. Our analysis instead probes the joint anchor-position and delay parameter space during self-calibration: re-parameterizing the eight anchor delays as a common mode plus regularized differentials ($d_i = c + e_i$), the fitted common mode is 112 mm and recovers metric scale from inter-anchor ranges alone (Sim(3) scale $0.958 \rightarrow 1.010$); clamping $c = 0$ raises the optimizer cost by about 70% while five perturbed initializations converge to the same common mode and scale.

We carry out a formal Fisher-information / profile-likelihood analysis of the coupling. Building the gauge-free Fisher information of the joint anchor-position/per-anchor-delay problem on the real Erlangen geometry — tag frames marginalized out by Schur complement, the six-dimensional SE(3) gauge removed — and reading the common-mode delay direction's marginal, the common delay is correlated with the isotropic layout-scale mode at $\rho = -0.977$, a variance inflation of $1/(1 - \rho^2) = 22\times$; equivalently, profiling the entire layout out of the common-delay parameter flattens its likelihood by $22\times$. In physical units the alias is 1 mm of common delay $\leftrightarrow -1.22$ mm of layout scale at the array edge (-645 ppm), with a profile-likelihood valley elongation of 8.9:1 (Figure 1). This confirms the delay–scale direction is *weakly observable yet still identifiable* — a strong near-degeneracy, not a strict one — consistent with both the practical cost-clamp indicator above and the recovered Sim(3) scale 0.958 (the AutoPos layout is expanded by $1/0.958 \approx 1.044$: an under-absorbed positive common-mode range bias lengthens every inter-anchor distance; in antenna-delay units the same effect carries the opposite sign, consistent with the 1 mm delay $\rightarrow -1.22$ mm edge-scale alias). One honest caveat: the globally-softest Fisher modes of this thin two-layer array are within-layer wiggle artifacts with near-zero overlap with every interpretable direction; the delay–scale coupling is correctly read from the delay direction's marginal, not from the softest raw eigenvector. The cost-clamp ratio remains only a practical corroborating indicator, not the identifiability statement.

We separately examined whether the *vertical* component of the static error is an observability deficit or a delay effect; this is a tag-side question, distinct from the anchor-side delay–scale direction discussed above. The eight-anchor two-layer array already has well-conditioned vertical

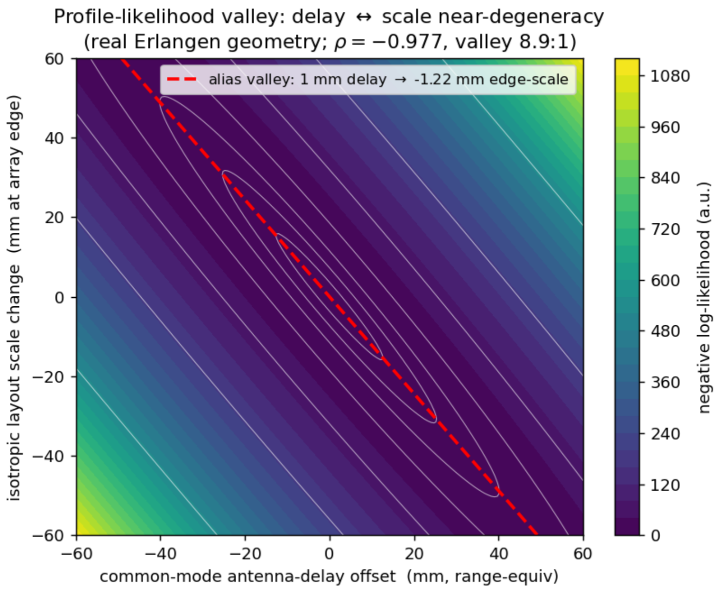

Figure 1: Profile-likelihood valley of the joint anchor self-calibration on the real Erlangen geometry, projected onto the (common-mode antenna-delay, isotropic layout-scale) plane (both axes in mm at the array edge). The long diagonal trough is the delay–scale near-degeneracy: a common-mode delay is absorbed almost entirely as a layout-scale change ($\rho = -0.977$, variance inflation $22\times$, alias slope 1 mm delay $\rightarrow -1.22$ mm edge-scale). This is the anchor-side mechanism behind the wrong-geometry-wins paradox.

geometry, so no observability deficit is expected; the self-calibration vertical-CRLB analysis (derived in the companion system paper: singular for a single coplanar layer, $\partial r/\partial y = 0$, falling to about 63 mm for the balanced four-plus-four array, essentially the production vertical error) confirms the vertical geometry is at its observability floor, so the residual is not a geometry deficit. Instead, because every tag observes the anchors with limited elevation diversity, a common-mode tag range bias aliases predominantly into the vertical coordinate: a geometric dilution-of-precision analysis of the deployment gives a vertical-to-horizontal alias ratio of about seven and VDOP/HDOP ≈ 1.6, the UWB analogue of the GPS receiver-clock/altitude coupling. As an independent empirical check, an active vertical-information injection – a tilted two-tag rotating arm sweeping the rotation plane from horizontal to vertical during self-calibration – did not reduce the recovered vertical error and showed no dose–response with rotation tilt under either a free-point or a rigid-body arm model, confirming the vertical error is dominated by this per-tag delay–altitude alias rather than by anything additional vertical calibration data could supply (see the supplementary experimental report).

This literature supports our Fisher/profile-likelihood language, but it does not remove the novelty. The Erlangen result is not just that a ranging network can be weakly identifiable. It is that a concrete range-only UWB self-calibration has a delay–layout direction whose in-environment winner changes when the tag-anchor positive range tail is reduced.

2.5 Evaluation Methodology

Several prior systems use ground truth, motion capture, or external trajectory sources for evaluation (Lutz et al., 2019; Ledergerber & D'Andrea, 2018; Queralta et al., 2020). What is less common is an explicit separation between anchor-side metric correctness and tag-side same-environment positioning accuracy. The Erlangen analysis uses Sim(3) scale, transfer matrices, winner's-curse correction, nested spatial validation, phase-center sensitivity, and raw-frame ranking reversal as separate checks. This is a measurement-validity protocol rather than a new real-time localization algorithm. All figures here derive from fixes (and an inter-anchor self-calibration) in which **all eight anchors of the 4+4 array participate**; the result is a property of that full-array participation, not of the ranging protocol—for a static fix the eight ranges are equivalent whether gathered sequentially or by broadcast. A reproduction that admits fixes from fewer anchors (e.g. a four-anchor subset) faces a much larger vertical CRLB or, for coplanar subsets, the singular regime, and is not expected to reproduce these numbers.

2.6 Independent Observation of the Wrong-Calibration-Wins Paradox

The delay–layout coupling documented in our Erlangen campaign is not unique to our setup. Luder et al. recently reported that self-localized UWB anchor positions produced better tag accuracy than laser-measured ground truth positions (13.96 cm versus 16.57 cm average 2D error) in an outdoor 600 m^2 deployment using DWM3000 hardware (Luder et al., 2025). The authors described the self-localized result as "more accurate" but did not investigate the underlying cause. Their observation is *consistent with* our delay–layout interpretation, but we do not present it as confirmation: the reported 13.96 versus 16.57 cm difference is within one standard deviation over only seven tag positions, and could equally arise from laser-survey error, 2D projection, anchor-height handling, or environment-specific NLOS. We therefore cite AniTrack as an independent external observation that motivates the question, not as a replication of the mechanism, and not as evidence that the coupling is a general property.

2.7 Generalization Beyond UWB

We note, as a discussion-level conjecture rather than a contribution, that the delay–layout coupling in this campaign resembles a broader identifiability pitfall in ranging-based self-calibration: where a per-device nuisance parameter (device delay, clock offset, sound speed) and a geometric parameter (layout scale) share a weakly observable direction, structured bias can make the metric-incorrect (scale-biased) solution appear empirically better. The analogy is imperfect and we flag its limit explicitly: antenna delay enters the range model *additively* (a constant offset) while layout scale is *multiplicative*, so the cancellation is only approximate over the deployment's working-distance range and would mis-track if distances changed substantially. We therefore treat GNSS clock–geometry and acoustic sound-speed–geometry as motivating cross-domain analogies, not as evidence of generality, and we make no cross-domain empirical claim.

2.8 Cross-Domain Prior Art on Scale–Propagation-Parameter Coupling

The scale–delay coupling identified in our campaign is not conceptually unique to UWB. In acoustic microphone array self-calibration, Burgess, Kuang, and Åström (Burgess et al., 2015) characterize the scale and offset ambiguities in TOA/TDOA self-calibration as affine-subspace rank deficiencies; Thrun (Thrun, 2005) recovers acoustic geometry only up to an affine (scale-bearing) transform; Kuang and Åström (Kuang & Åström, 2013) solve transmitter/receiver geometry up to an unknown affine map before a metric upgrade; and Crocco et al. (Crocco et al., 2012) jointly self-calibrate sensor positions and timing offsets. Critically for our framing, these works treat delay/offset as a *first-class jointly-solved unknown with formally characterized degeneracies*, not as a nuisance correction—so our claim is not that delay-as-unknown is new, but that the UWB additive-delay *in-environment cancellation* (where the absorbed error helps positioning in the deployment environment) differs from their affine/dimensional degeneracies. Within UWB, Batstone et al. (Batstone et al., 2017) already perform range-only TOA anchor self-calibration, and recent UWB-inertial work that augments per-anchor multiplicative scale and additive bias states jointly (e.g. MR-ULINS, arXiv:2408.05719) is the closest parameterization to ours—though it does not characterize the scale–delay pair as a weakly observable direction. In cooperative wireless network localization, Shen, Wymeersch, and Win (Shen et al., 2010) show that without sufficient anchors the Fisher information matrix is rank-deficient due to geometric and clock-related ambiguities. In GNSS, Pan et al. (Pan et al., 2015) demonstrate that satellite-clock estimation can absorb part of the orbit error, and this error absorption *improves* positioning accuracy by 38.8% and 36.1% in their rover tests, making it the closest cross-domain analog to "error absorption helps positioning."

These works, together with the textbook bias–variance / shrinkage principle that a deliberately biased model can lower prediction error (Hastie et al., 2009), establish that the *existence* of a scale–propagation-parameter coupling and of useful-bias effects is prior art. Our contribution is not the bare existence of the coupling, but its specific manifestation in UWB anchor self-calibration: (i) the weakly observable direction between per-device antenna delay and global metric scale, indicated by common-mode scale recovery and a cost-clamp test, now quantified by a gauge-free Fisher/profile-likelihood analysis ($\rho = -0.977$, 22$\times$ variance inflation); (ii) the result that this coupling makes the scale-biased layout *outperform* the metric-correct one on same-environment positioning when the tag delay is uncorrected—a UWB manifestation we did not find explicitly reported, while acknowledging GNSS clock–orbit absorption (Pan et al., 2015) and acoustic scale ambiguity (Burgess et al., 2015) as closely related cross-domain precedents; and (iii) the use of structured-range-bias reduction as a diagnostic intervention to reveal the true calibration quality.

3 Why the Coupling Has Not Been Identified Before

The UWB positioning literature divides into two groups that rarely separate metric correctness from in-environment accuracy, which is why delay–layout coupling is easy to miss. We do not claim it *cannot* be observed—only that the standard evaluations do not surface it.

Group 1: Anchor positions assumed known. The majority of UWB positioning papers treat anchor coordinates as pre-surveyed infrastructure (Prorok, 2013; Wymeersch et al., 2012; Kong, 2023; Sung et al., 2023). In this paradigm, geometry is fixed and delay is calibrated independently against known distances. The two parameter spaces never interact in the same estimation problem. Any delay–geometry coupling is invisible because there is no geometry to couple with: the anchor positions are inputs, not outputs.

Group 2: Anchor positions estimated. A smaller body of work estimates anchor positions from inter-anchor ranges, robot trajectories, or auxiliary sensing (Hamer & D'Andrea, 2018; Ridolfi et al., 2021; Van Herbruggen et al., 2023; Piavanini et al., 2022). These papers jointly estimate positions and sometimes delays, so the coupling can in principle arise. However, they evaluate success by *tag positioning accuracy*, not by *metric correctness of the anchor geometry*. If the coupling produces a scale-biased layout that cancels NLOS bias and gives low tag error, it is reported as a successful calibration. Corbalan et al. (Corbalan et al., 2023) are a partial exception—they do compare self-localized geometry against ground truth and report that geometry error propagates only weakly to tag error—but they do not construct a case where the scale-biased layout *wins*, nor isolate a bias-cancellation cause. The AniTrack result (Luder et al., 2025) is a concrete example: self-localized anchors outperformed ground truth anchors on tag accuracy, and the authors reported this as a positive outcome without investigating the geometry.

What is needed to observe the coupling. Three conditions must be met simultaneously: (1) range-only anchor self-calibration, so that delay and geometry can interact in the solver; (2) external ground truth for the anchor geometry, so that metric correctness can be evaluated independently of tag accuracy; and (3) comparison of multiple solver parameterizations, so that the ranking paradox becomes visible. The Erlangen campaign satisfies all three. Most prior work satisfies at most one.

4 Novelty Gap Table

Table 1: Novelty gaps after full-text checking.

Topic	Prior art covers	Remaining gap addressed here
UWB anchor self-calibration	Automatic anchor localization, robot-aided calibration, multi-hop calibration.	Same-environment accuracy can prefer a scale-biased layout.
Antenna delay calibration	Device delay, received-power bias, module-dependent range models.	Delay error is analyzed as coupled to layout scale, not only as a standalone nuisance.
NLOS mitigation	Classification, weighting, CIR/statistical features, learned corrections.	Range-bias reduction is used as an intervention that reverses the calibration ranking (reported separately).
Identifiability	Graph rigidity, observability, Fisher/CRB bounds.	The weak direction is interpreted physically as delay–layout coupling in a measured UWB campaign.
Evaluation	Ground truth and held-out experiments appear in individual systems.	Metric correctness, p50 accuracy, post-selection optimism, and dynamic transfer are separated explicitly.
Independent paradox observation	Self-localized anchors can sometimes outperform surveyed anchors in tag accuracy.	Why this happens, and whether it indicates a calibration quality problem rather than a calibration success.
Researcher-group split	UWB research divides into known-anchor positioning and self-calibration, evaluated by different criteria.	Neither group checks whether self-calibrated geometry is metrically correct and whether tag accuracy is a reliable proxy for calibration quality.
Cross-domain coupling	Scale–propagation-parameter coupling is known in acoustic and GNSS self-calibration, and GNSS error absorption can improve positioning.	The coupling interacts with structured range bias to make the scale-biased layout win in-environment, which has not been reported in prior ranging systems.
Generality	Other ranging systems know clock, bias, and geometry couplings.	The paper shows a concrete scale-bias cancellation failure mode (a scale-biased layout winning in-environment under an uncorrected tag delay) with Vicon-grounded evidence.

5 Our Contribution

This paper makes two contributions that together form its headline: a Vicon-grounded *quantitative characterization* of the delay–layout coupling (we do not claim first discovery of the qualitative effect), and a methodological *falsification protocol* built on it. We foreground the two jointly rather than ranking one above the other.

The first contribution is empirical. The qualitative observation that a common delay offset trades off with geometric scale in ranging self-calibration has been noted in acoustic/TOA array self-calibration (Burgess et al., 2015); in UWB, the closely related self-calibration of Hamer & D'Andrea (2018) jointly handles clock synchronization and anchor-position estimation. We provide an explicit, Vicon-validated *quantitative characterization* of this coupling in UWB anchor self-calibration (we do not claim to be first to observe it, given the cross-domain and AniTrack precedents above). A four-way ablation that crosses two layout sources (self-calibrated `v4-io` coordinates, Vicon-measured coordinates) with residual-correction treatments shows the fitted corrections are frame-conditioned: Vicon-measured anchor coordinates give 311 mm RMSE with no correction and 252 mm with transplanted corrections, but 78 mm once the corrections are re-estimated in the Vicon frame. A common-mode delay re-parameterization ($d_i = c + e_i$, $c = 112$ mm) recovers metric scale from inter-anchor ranges alone, moving the AutoPos-to-Vicon Sim(3) scale from 0.958 to 1.010 and the rigid anchor RMSE from 105.4 to 63.0 mm; clamping the common mode to zero raises the solver cost by about 70% and five perturbed initializations converge to the same scale, indicating a weakly observable—not strictly degenerate—delay–scale direction (now confirmed by the gauge-free Fisher analysis above: common-mode delay correlates with isotropic layout scale at $\rho = -0.977$, a 22× variance inflation). We then show that the metric-correct common-mode layout is the worse same-environment positioner (109.5 mm median) than the scale-biased `v4-io` layout (72.7 mm) *only when the tag delay is left uncorrected*: the scale-biased layout's scale error compensates an uncorrected positive tag-device delay. The fair-comparison control points the same way: the physically honest correction is a *single global* tag-delay constant, because the deployed custom SS-TWR firmware applies one generic antenna-delay default (the vendor example value, 16436 device units) identically to every node—all eight anchors and the tag—with no per-device calibration, so the residual per-device delays are exactly what the self-calibration absorbs into geometry. With a leave-one-out, no-peek constant ($\approx$ 50 mm, the median ranging residual on held-out positions) the metric-correct layout becomes the better positioner: 68.5 mm held-out median (P95 156 mm) versus 72.7 mm. The effect is carried by the paired comparison and the tail, not by the marginal-median gap — the ~4 mm median difference has a paired bootstrap confidence interval spanning zero, but across the 24 static positions the metric-correct layout is better at 17 (paired signed-rank $p \approx 0.02$) and sharply reduces the error tail (RMSE 109.8 to 82.8 mm; P95 171.5 to 156 mm), whereas per-frame joint estimation only ties the median (72.9 mm) by overfitting and blows the tail to P95 317.5 mm. This closes the causal chain: the scale-biased layout's apparent advantage was an uncorrected constant the wrong geometry happened to absorb, not a genuine superiority.

The deeper reading: self-consistent is not metric. Same-environment accuracy cannot certify a calibration because the self-calibrated layout is internally *self-consistent* but not *metric*. The common-mode delay and the isotropic layout scale span a near-degenerate direction ($\rho = -0.977$), so a whole family of (scale, delay) solutions reproduces the inter-anchor ranges equally well; the range data alone cannot select the metrically true member, and `v4-io` is simply the member the solver happened to pick, its 4.4% scale bias silently encoding the uncorrected common-mode tag delay. It positions well only *inside its own delay convention*: correcting the geometry without also correcting the delay steps off the self-consistent manifold, and the apparent advantage disappears. Selecting the metric member therefore requires information outside the ranges—one known baseline, the factory common-mode delay, or an external metric reference such as Vicon—to pin the degenerate scale direction. This is also why the transfer claim must be earned rather than assumed: sweeping the layout across rooms is precisely the lever that exercises the degenerate direction and forces the true scale to reveal itself.

Factory OTP does not supply the per-device delays. A natural objection is that the

DWM1001C stores a factory antenna-delay calibration in DW1000 OTP, so applying it should remove the per-device delay the coupling exploits. We read the OTP of all nine modules (eight anchors and the static tag, one manufacturing lot) and verified the lone outlier with five repeated power-cycle reads. The stored antenna delay is *near-uniform* — eight modules read 16472 device units and one anchor (confirmed stable) reads 16451 — and it does *not* track the per-device delay the solver must recover: the seven modules sharing the identical OTP value have fitted delay terms spanning roughly 49–148 mm, and the single genuine outlier's fitted term matches that of a same-OTP module rather than reflecting its 21-unit (≈ 98 mm) OTP offset. The per-device antenna-calibration-temperature field is unprogrammed on every unit, indicating no per-device antenna-delay calibration was performed — although the same memory does carry per-device crystal-trim and temperature-sensor references that vary unit to unit. Applying the OTP value is therefore a near-uniform common-mode shift that only rescales the layout and cannot supply the per-device delay differences that drive the coupling. The per-device delays are thus not recoverable from factory data — only from per-unit bench calibration (not done in deployment) or self-calibration, which entangles them with geometry — the effect studied here. The coupling is intrinsic to these commodity single-antenna modules (DWM1001C/DW1000), not an artifact of a skipped calibration step.

The evidence chain is falsification-aware. The common-mode parameterization absorbs bulk anchor delay and fixes scale, and the per-anchor range-bias decomposition closes the error budget: eight positive per-device terms with mean 94.6 mm (anchor A largest at 148.2 mm) reproduce the measured pairwise excess to within 0.11 mm. Phase-center sensitivity shows that the `v4-io`-over-common-mode static ranking is robust to marker-offset perturbations beyond 10 mm, while the Vicon-oracle rank is fragile near 2 mm and is therefore not used as the sole proof. The raw-frame intervention that reduces the positive tag-anchor tail reverses the in-environment ranking in favour of the metric-correct layout, providing the causal evidence that the scale-biased layout's advantage came from cancellation rather than from a generally better physical calibration.

The second contribution is methodological. We provide a Vicon-grounded error-analysis protocol that separates metric correctness, same-environment positioning accuracy, post-selection optimism, and dynamic transfer. The protocol includes Sim(3) scale validation against Vicon, a transfer matrix over layout and delay choices, winner's-curse correction, nested spatial cross-validation, phase-center sensitivity analysis, NLOS leakage auditing, and ROTO gap decomposition. This matters because a single-number accuracy report would have called the `v4-io` layout the winner and hidden the physical reason—and indeed, the AniTrack result (Luder et al., 2025) shows exactly this: without the falsification protocol, the scale-bias cancellation phenomenon (a scale-biased layout winning in-environment) is reported as a success rather than diagnosed as a coupling artifact.

Priority, disclosure, and reproducibility. We stake the flag not on the *existence* of the delay–layout coupling — the acoustic, GNSS, and AniTrack precedents above forbid that claim — but on three things later UWB self-calibration work cannot route around. (i) The first gauge-free Fisher/profile-likelihood *quantification* of the delay$\leftrightarrow$layout-scale near-degeneracy on real UWB hardware: common-mode delay correlated with isotropic layout scale at $\rho = -0.977$, a $22\times$ variance inflation, an alias of 1 mm delay $\leftrightarrow$ -1.22 mm edge-scale (-645 ppm), with profile-likelihood valley elongation 8.9:1. (ii) The conditional wrong-geometry-wins regime *and* the control that closes it: a single global tag-delay constant flips the in-environment ranking so the metric-correct layout becomes the better positioner at most static positions (68.5 vs 72.7 mm median; better at 17 of 24, paired $p \approx 0.02$, with the error tail cut from 171.5 to 156 mm P95), establishing that the scale-biased layout's advantage was an uncorrected constant, not a better calibration. (iii) The Vicon-grounded falsification protocol that separates metric correctness from same-environment accuracy. For defensive-publication value the manuscript states the enabling combination once — range-only UWB anchor self-calibration + independent metric (Vicon) ground truth for the *geometry* + multiple solver parameterizations compared on that ground truth — the exact three-condition conjunction that prior work satisfies at most one of (§"What is needed to observe the coupling"). The Fisher derivation and figure ship as a runnable artifact, making the $\rho = -0.977$ / 22× numbers and the $0.958 \to 1.010$ scale recovery independently reproducible: a reusable artifact later UWB self-calibration evaluations must cite to claim a self-calibrated layout is metrically correct, not merely a good in-environment positioner. *Citable priority sentence: "In range-only UWB anchor self-calibration the common-mode antenna-delay parameter is weakly observable against isotropic layout scale — measured at $\rho = -0.977$ (22× variance inflation) on the deployed eight-anchor array — so same-environment tag accuracy can rank a scale-biased layout above a metric-correct one until a single global delay constant is applied — after which the metric-correct layout becomes the better positioner."*

6 Nearest Prior Art

Table 2: Closest papers and specific differences.

Paper	What they specifically did	How our work differs	Why the difference matters
Lutz et al. (2019)	Used visual-inertial SLAM and fiducial markers to estimate UWB anchor poses and sensor error-model parameters.	We analyze range-only self-calibration where delay and layout scale are weakly coupled, without using VINS as the calibration truth source.	Their method improves calibration with external visual structure; ours explains why a radio-only calibration can look better while being metric-incorrect (scale-biased).
Hamer & D'Andrea (2018)	Built a self-calibrating one-way UWB network with clock synchronization and anchor position computation for multi-robot localization.	We focus on validation against Vicon and on the layout/delay/range-bias cancellation that lets a scale-biased self-calibrated layout outperform the metric-correct one in-environment.	A working self-calibrating network does not by itself tell whether the empirically best layout is physically correct.
Luder et al. (2025) (AniTrack)	Deployed a self-localizing UWB system for animal tracking and found self-localized anchors outperformed laser-surveyed anchors in tag accuracy (13.96 vs 16.57 cm average error).	We explain why this can happen: delay–layout coupling lets a scale-biased self-calibrated layout cancel structured range bias. We validate with Sim(3) scale, a four-way layout/delay ablation, and common-mode scale recovery.	Their unexplained observation is consistent with our interpretation but is not a replication: the 13.96 vs 16.57 cm gap is within one standard deviation over seven positions and has competing explanations (survey error, 2D projection). We propose and Vicon-test a candidate mechanism.

Paper	What they specifically did	How our work differs	Why the difference matters
Pan et al. (2015) (GNSS)	Showed satellite-clock estimation absorbs orbit error, improving PPP positioning accuracy and convergence.	Their error absorption is clock-absorbs-orbit in GNSS, where satellite infrastructure is not self-calibrated by the rover. Our coupling is between self-calibrated anchor geometry and per-device delay, where a scale-biased layout cancels structured positive range bias.	They demonstrate that coupled-error absorption can help positioning. We show that it can make a metric-incorrect calibration win over the correct one when the tag bias is uncorrected.
Ledergerber & D'Andrea (2018)	Calibrated module-dependent delays and range inaccuracies by maximum likelihood, without external ground truth.	We show that improving a delay/range model is not the same as resolving scale coupling in anchor self-calibration.	Their result motivates range calibration; our result warns that calibration criteria can reward the wrong geometric solution.
Shalaby et al. (2023)	Proposed scalable antenna-delay calibration and received-power-dependent bias/uncertainty models.	We do not propose another delay estimator; we show delay estimates can interact with layout scale and structured NLOS.	It reframes delay from a standalone correction into a parameter that can hide a geometric error.
De Preter et al. (2019)	Modeled range bias and autocalibration in a UWB positioning system.	We test whether reducing structured range bias changes the calibration ranking, rather than only improving a biased range model.	This turns range-bias reduction into evidence about calibration validity.
Di Franco et al. (2017)	Localized a network from biased, multimodal NLOS UWB measurements without prior calibration.	We calibrate and compare physical layouts, including a metric-correct common-mode layout and a scale-biased `v4-io` layout.	Their NLOS robustness is not a test of whether a scale-biased layout can win under biased tag ranges.

7 What We Are Not Claiming

8 Target Venue and Titles

The primary target venue remains *Measurement*, because the paper is a metrology and validation study: it asks whether the calibration that gives the best positioning number is physically correct. IEEE *Transactions on Instrumentation and Measurement* is a natural backup because the evidence chain involves calibration, uncertainty, ground-truth comparison, and measurement validity.

Table 3: Claims excluded from the manuscript narrative.

Not claimed	Reason
CIR-free deployable NLOS classifier	The raw-frame lower-tail statistic is a separately reported static batch intervention, not a real-time classifier.
Real-time dynamic improvement	ROTO has one-frame ranges and remains around 101.5 mm under the conservative alignment.
Common-mode layout transfers better to new rooms	The scale argument suggests transfer, but an independent room is still needed.
Mixture-model superiority	The simple lower-tail statistic beat the tested mixtures in this dataset (reported separately).
20–30 mm UWB accuracy	The static median floor remains tens of millimetres and the P95 tail remains high.
Vicon oracle proves cancellation alone	Phase-center sensitivity makes that single argument fragile.
Rigid-body ROTO improvement	The tested two-tag rigid solvers worsened the dynamic result.

Table 4: Candidate titles.

Rank	Title
1	Metric Correctness versus Positioning Accuracy in UWB Anchor Self-Calibration: A Vicon-Grounded Falsification Study
2	When Delay Absorbs Geometry: A Conditional Wrong-Geometry-Wins Failure Mode in UWB Self-Calibration
3	Delay–Layout Coupling in UWB Self-Calibration: Why Same-Environment Accuracy Can Reward a Distorted Layout

9 Risk Assessment

The strongest publication risk is overclaiming. The literature search confirms that UWB self-calibration, antenna-delay calibration, range-bias modeling, and NLOS mitigation are well studied. The manuscript must therefore emphasize the in-environment cancellation result and the frame-conditioned residual corrections, not a raw accuracy number. The second risk is external validity. The Erlangen campaign is one room with 24 static positions, so common-mode-layout transfer should remain an expectation until frozen-pipeline second- and N-room validation is complete. The third risk is tail behavior: the metric-correct layout improves the median under joint tag-delay estimation but degrades the tail (P95 317.5 mm), so the paper should separate median mechanism evidence from worst-case performance.

A fourth risk is that a reviewer familiar with acoustic self-calibration or GNSS may view the scale–delay coupling as a known ambiguity class and dismiss the contribution as incremental. The defence is that the *existence* of the coupling is not the claimed novelty. The novelty is the explicit Vicon-grounded characterization in UWB self-calibration, the demonstration that the coupling can make the scale-biased layout win in-environment under an uncorrected tag bias, and the diagnostic intervention that confirms cancellation. We did not find these elements combined in the acoustic or GNSS prior art, though we cite that work as closely related. The paper should preemptively cite the cross-domain works and draw the distinction explicitly.

References

Almansa, C.M., Shule, W., Queralta, J.P. and Westerlund, T., 2020. Autocalibration of a mobile UWB localization system for ad-hoc multi-robot deployments in GNSS-denied environments. *arXiv preprint arXiv:2004.06762.*

Aspnes, J., Eren, T., Goldenberg, D.K., Morse, A.S., Whiteley, W., Yang, Y.R., Anderson, B.D.O. and Belhumeur, P.N., 2006. A theory of network localization. *IEEE Transactions on Mobile Computing*, 5(12), pp.1663–1678.

Batstone, K., Oskarsson, M. and Åström, K., 2017. Towards real-time time-of-arrival self-calibration using ultra-wideband anchors. In: *2017 International Conference on Indoor Positioning and Indoor Navigation (IPIN).* IEEE, pp.1–8. DOI: 10.1109/IPIN.2017.8115885.

Burgess, S., Kuang, Y. and Åström, K., 2015. TOA sensor network self-calibration for receiver and transmitter spaces with difference in dimension. *Signal Processing*, 107, pp.33–42.

Corbalan, P., Picco, G.P., Coors, M. and Jain, V., 2023. Self-localization of ultra-wideband anchors: From theory to practice. *IEEE Access*, 11, pp.29711–29725.

Crocco, M., Del Bue, A. and Murino, V., 2012. A bilinear approach to the position self-calibration of multiple sensors. *IEEE Transactions on Signal Processing*, 60(2), pp.660–673.

De Preter, A., Goysens, G., Anthonis, J., Swevers, J. and Pipeleers, G., 2019. Range bias modeling and autocalibration of a UWB positioning system. In: *International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, pp.1–8.

Di Franco, C., Prorok, A., Atanasov, N., Kempke, B., Dutta, P., Kumar, V. and Pappas, G.J., 2017. Calibration-free network localization using non-line-of-sight UWB measurements. In: *ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN).*

Goudar, A., Zhao, W. and Schoellig, A.P., 2021. Online spatio-temporal calibration of tightly-coupled ultrawideband-aided inertial localization. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).*

Guvenc, I., Chong, C.-C., Watanabe, F. and Inamura, H., 2007. NLOS identification and weighted least-squares localization for UWB systems using multipath channel statistics. *EURASIP Journal on Advances in Signal Processing*, 2007.

Hastie, T., Tibshirani, R. and Friedman, J., 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction.* 2nd ed. Springer.

Hamer, M. and D'Andrea, R., 2018. Self-calibrating ultra-wideband network supporting multi-robot localization. *IEEE Access*, 6, pp.22292–22304.

Van Herbruggen, B., Luchie, S., Fontaine, J. and De Poorter, E., 2023. Multihop self-calibration algorithm for ultra-wideband anchor node positioning. *IEEE Journal of Indoor and Seamless Positioning and Navigation.*

Hesch, J.A., Kottas, D.G., Bowman, S.L. and Roumeliotis, S.I., 2014. Camera-IMU-based localization: Observability analysis and consistency improvement. *The International Journal of Robotics Research*, 33(1), pp.182–201.

Jeong, S.-G., Do, Q.-V., Hwang, H.-J., Hasegawa, M., Sekiya, H. and Hwang, W.-J., 2023. Hybrid quantum convolutional neural networks for UWB signal classification. *IEEE Access.*

Jourdan, D.B., Dardari, D. and Win, M.Z., 2008. Position error bound for UWB localization in dense cluttered environments. *IEEE Transactions on Aerospace and Electronic Systems*, 44(2), pp.613–628.

Kong, Q., 2023. NLOS identification for UWB positioning based on IDBO and convolutional neural networks. *IEEE Access.*

Kuang, Y. and Åström, K., 2013. Stratified sensor network self-calibration from TDOA measurements. In: *21st European Signal Processing Conference (EUSIPCO).*

Ledergerber, A., Hamer, M. and D'Andrea, R., 2015. A robot self-localization system using one-way ultra-wideband communication. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp.3131–3137.

Ledergerber, A. and D'Andrea, R., 2018. Calibrating away inaccuracies in ultra wideband range measurements: A maximum likelihood approach. *IEEE Access*, 6, pp.78719–78730.

Liu, Z., Hakala, T., Hyyppa, J., Kukko, A., Kaartinen, H. and Chen, R., 2024. Data-driven antenna delay calibration for UWB devices for network positioning. *IEEE Transactions on Instrumentation and Measurement.*

Lutz, P., Schuster, M.J. and Steidle, F., 2019. Visual-inertial SLAM aided estimation of anchor poses and sensor error model parameters of UWB radio modules. In: *International Conference on Advanced Robotics (ICAR)*, pp.739–746.

Luder, V., Schulthess, L., Cortesi, S., Rivero Davis, L. and Magno, M., 2025. AniTrack: A power-efficient, time-slotted and robust UWB localization system for animal tracking in a controlled setting. *arXiv preprint arXiv:2506.00216.*

Marano, S., Gifford, W.M., Wymeersch, H. and Win, M.Z., 2010. NLOS identification and mitigation for localization based on UWB experimental data. *IEEE Journal on Selected Areas in Communications.*

Meghani, S.K., Asif, M., Awin, F. and Tepe, K., 2019. Empirical based ranging error mitigation in IR-UWB: A fuzzy approach. *IEEE Access.*

Pan, S., Chen, W., Jin, X., Shi, X. and He, F., 2015. Real-time PPP based on the coupling estimation of clock bias and orbit error with broadcast ephemeris. *Sensors*, 15(7), pp.17808–17826.

Patwari, N., Ash, J.N., Kyperountas, S., Hero, A.O., Moses, R.L. and Correal, N.S., 2005. Locating the nodes: Cooperative localization in wireless sensor networks. *IEEE Signal Processing Magazine*, 22(4), pp.54–69.

Piavanini, M., Barbieri, L., Brambilla, M., Cerutti, M., Ercoli, S., Agili, A. and Nicoli, M., 2022. A calibration method for antenna delay estimation and anchor self-localization in UWB systems. In: *IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0&IoT)*, pp.173–177.

Prorok, A., 2013. *Models and Algorithms for Ultra-Wideband Localization in Single- and Multi-Robot Systems.* PhD thesis. EPFL.

Queralta, J.P., Almansa, C.M., Schiano, F., Floreano, D. and Westerlund, T., 2020. UWB-based system for UAV localization in GNSS-denied environments. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).*

Ridolfi, M., Kaya, A., Berkvens, R., Weyn, M., Joseph, W. and De Poorter, E., 2021. Self-calibration and collaborative localization for UWB positioning systems. *ACM Computing Surveys.*

Shah, S., Chaiwong, K., Kovavisaruch, L., Kaemarungsi, K. and Demeechai, T., 2021. Antenna delay calibration of UWB nodes. *IEEE Access*, 9, pp.63294–63305.

Shah, S., Kovavisaruch, L., Kaemarungsi, K. and Demeechai, T., 2022. Node calibration in UWB-based RTLSs using multiple simultaneous ranging. *Sensors*, 22(3), 864.

Shalaby, M., Cossette, C.C., Forbes, J.R. and Le Ny, J., 2023. Calibration and uncertainty characterization for ultra-wideband two-way-ranging measurements. In: *IEEE International Conference on Robotics and Automation (ICRA).*

Shen, Y., Wymeersch, H. and Win, M.Z., 2010. Fundamental limits of wideband localization—Part II: Cooperative networks. *IEEE Transactions on Information Theory*, 56(10), pp.4981–5000.

Shi, Q., Zhao, S., Cui, X., Lu, M. and Jia, M., 2019. Anchor self-localization algorithm based on UWB ranging and inertial measurements. *Tsinghua Science and Technology.*

Sung, S., Kim, H. and Jung, J.-i., 2023. Accurate indoor positioning for UWB-based personal devices using deep learning. *IEEE Access.*

Thrun, S., 2005. Affine structure from sound. In: *Advances in Neural Information Processing Systems (NIPS)*, 18.

Venkatesh, S. and Buehrer, R.M., 2007. Non-line-of-sight identification in ultra-wideband systems based on received signal statistics. *IET Microwaves, Antennas & Propagation.*

Wymeersch, H., Lien, J. and Win, M.Z., 2009. Cooperative localization in wireless networks. *Proceedings of the IEEE.*

Wymeersch, H., Marano, S., Gifford, W.M. and Win, M.Z., 2012. A machine learning approach to ranging error mitigation for UWB localization. *IEEE Transactions on Communications.*

Yang, H., Wang, Y., Seow, C., Sun, M. and Plets, D., 2024. UWB NLOS identification and mitigation based on bidirectional encoder representations from transformer (BERT) deep learning. In: *International Conference on Indoor Positioning and Indoor Navigation (IPIN).*

Yogesh, V., Grevinga, L., Voort, C., Buurke, J.H., Veltink, P.H. and Baten, C.T.M., 2024. Novel calibration method for improved UWB sensor distance measurement in the context of application for 3D analysis of human movement. *Engineering Science and Technology, an International Journal*, 58, 101844.